Design Brief & Methodology Statement Project: TD Term Deposit Campaign Performance Dashboard **Author:** Manish Arora **Date:** January 7, 2026

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1. Executive Summary

Goal The objective of this analysis was to optimize the effectiveness of Term Deposit marketing campaigns by identifying high-conversion customer segments and eliminating operational inefficiencies. The resulting dashboard provides a strategic view for executives to reallocate budget toward high-probability targets.

Key Findings

- **Targeting Strategy:** The analysis identified two hyper-performing segments: **Students (29% conversion)** and **Retirees (23% conversion)**. These groups convert at 2x-3x the rate of the general population (11.7%), suggesting that future campaigns should prioritize age-based and occupation-based targeting.
- **Seasonal Timing:** Campaign performance is highly seasonal, with significant peaks in **March and September**. Conversely, high-volume outreach during the summer months (May and July) yielded the lowest returns, indicating a need to reduce summer spend.
- **Operational Efficiency:** A "Diminishing Returns" analysis revealed that conversion probability drops below 10% after the **4th call attempt**. Continuing outreach beyond this point yields negative ROI.

Strategic Recommendations

1. **Implement a "4-Call Cap":** Cease outreach after four unsuccessful attempts to reduce operational costs by approximately 30% without significantly impacting total sales.
2. **Shift Budget:** Reallocate resources from low-performing summer months to aggressive targeted campaigns in March and September.
3. **Cross-Sell Focus:** Prioritise "Debt-Free" customers, as data shows that existing loan holders have a significantly lower propensity to save.

2. Data Logic & Assumptions

To ensure data integrity and user clarity, the following transformation steps were applied to the raw dataset:

- **Data Cleaning & Terminology:** The raw field y (target variable) was renamed to **"Outcome"**, and values were mapped from "yes/no" to **"Subscribed / Not Subscribed"**. This aligns the dashboard with standard banking terminology.
- **Handling Ambiguity:** Records with "Unknown" values in key demographic fields (Job, Education) were filtered out of top-level visuals to prevent skewed insights, while ensuring the overall success rate calculation remained accurate across the total population.
- **Metric Definition: Conversion Rate** was calculated as the simple average of successful outcomes (Count of Subscribed/ Total Rows) Given the granular nature of the dataset, a weighted average was not required.

3. Strategic Modeling (The "Why")

Beyond standard descriptive analytics, two strategic models were developed to answer the "So What?" for the business:

A. Call Frequency & Efficiency Modeling

- **Problem:** Call centers often waste resources on "stubborn" leads.
- **Analysis:** I plotted conversion rates against the Campaign (contact frequency) variable. The data revealed a sharp efficiency curve where success rates stabilize and then decline after the 4th contact.
- **Solution:** I implemented a visual "Stop Point" on the dashboard to guide operational managers on exactly when to stop spending money on a lead.

B. Financial Profile Segmentation

- **Problem:** Evaluating customers based on single product holdings (e.g., just "Housing Loan") failed to capture their total liquidity.
- **Analysis:** I created a composite "Financial Profile" variable that categorizes customers based on their full debt portfolio (Housing + Personal Loans).
- **Insight:** The model proved that "Debt-Free" customers are the primary drivers of Term Deposit sales, while "Dual Loan Holders" are high-risk targets for savings products.

4. Technical Appendix

This dashboard leverages advanced Power BI features to enhance storytelling and user experience:

Power Query (M Code) Custom logic was used to derive the *Financial Profile* column, simplifying complex cross-product logic into a single readable attribute:

Advanced DAX & Visualization

- **Dynamic Formatting:** Created custom DAX measures visually split the "Frequency" line chart. This visual technique subconsciously reinforces the "Good vs. Bad" operational zones for the user.